INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Governance, Risk and Compliance (GRC)

Week 2

Lab 2: Linux Scripting for GRC Compliance

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Date: 24/01/2026

Linux Scripting for GRC Compliance

Executive Summary

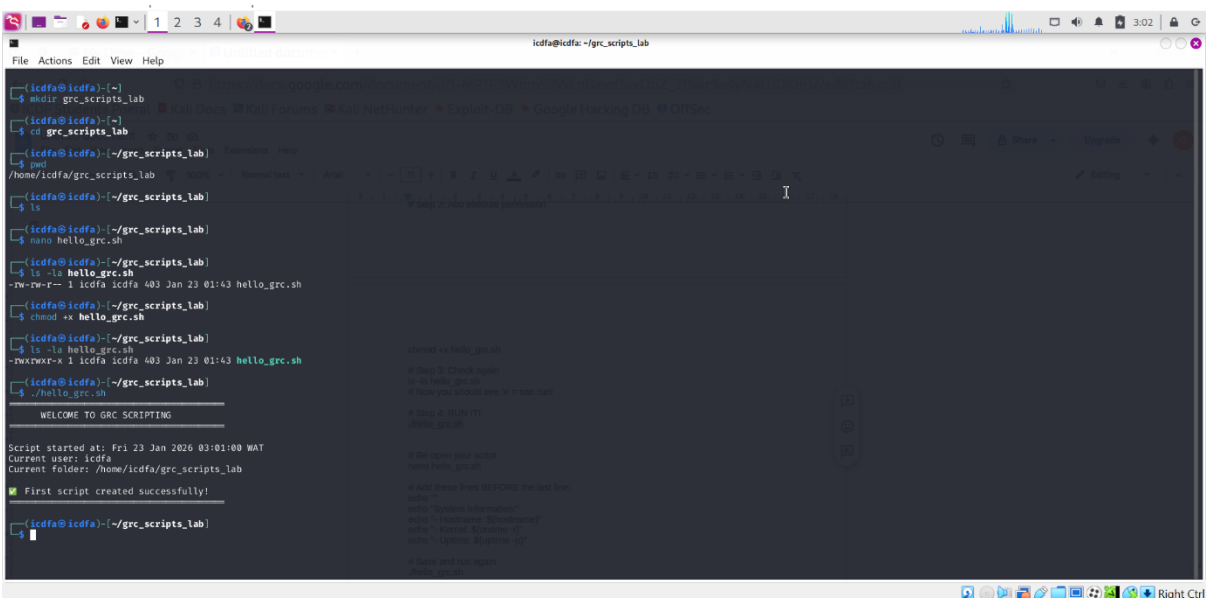
This report documents the practical knowledge and skills gained through the comprehensive "Linux Scripting for GRC Compliance" laboratory exercises. Over six structured sessions, I progressed from understanding basic Bash scripting operations to applying the skills to compliance automation frameworks. This hands-on approach helped me to understand and appreciate how to apply these technical skills to translate Governance, Risk, and Compliance (GRC) concepts into practical automated solutions, enhancing my competencies in compliance validation, risk assessment, and audit support through scripting.

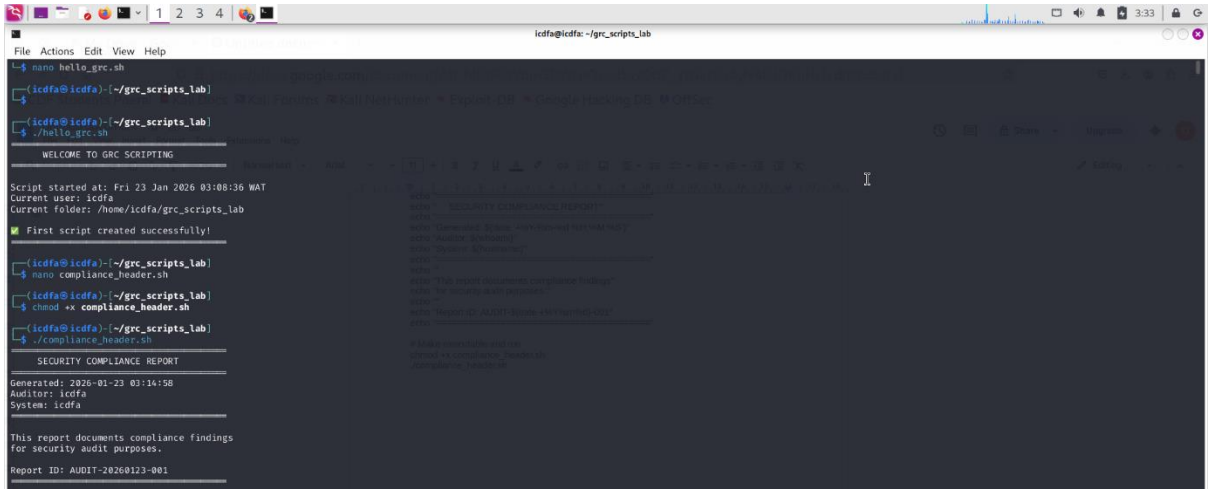
SESSION 1: ABSOLUTE BASICS - Your First Script

During this session, I was able to understand foundational scripting capabilities through practising the exercises.

Beginning with terminal navigation using *pwd* and *ls* commands, I progressed to creating my first executable script with nano hello_grc.sh. I learned how to create my first bash scripts and how to run and execute them. (chmod +x), (./script.sh)

These exercises established essential scripting patterns that formed the foundation for all subsequent GRC automation development, especially the shebang (#!/bin/bash).



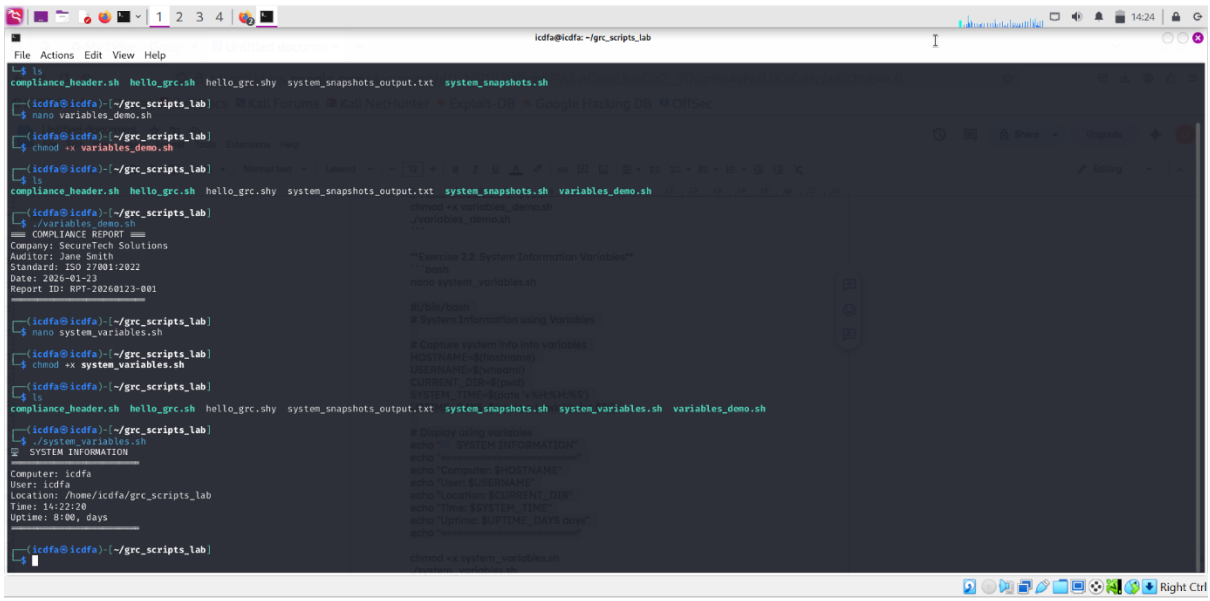


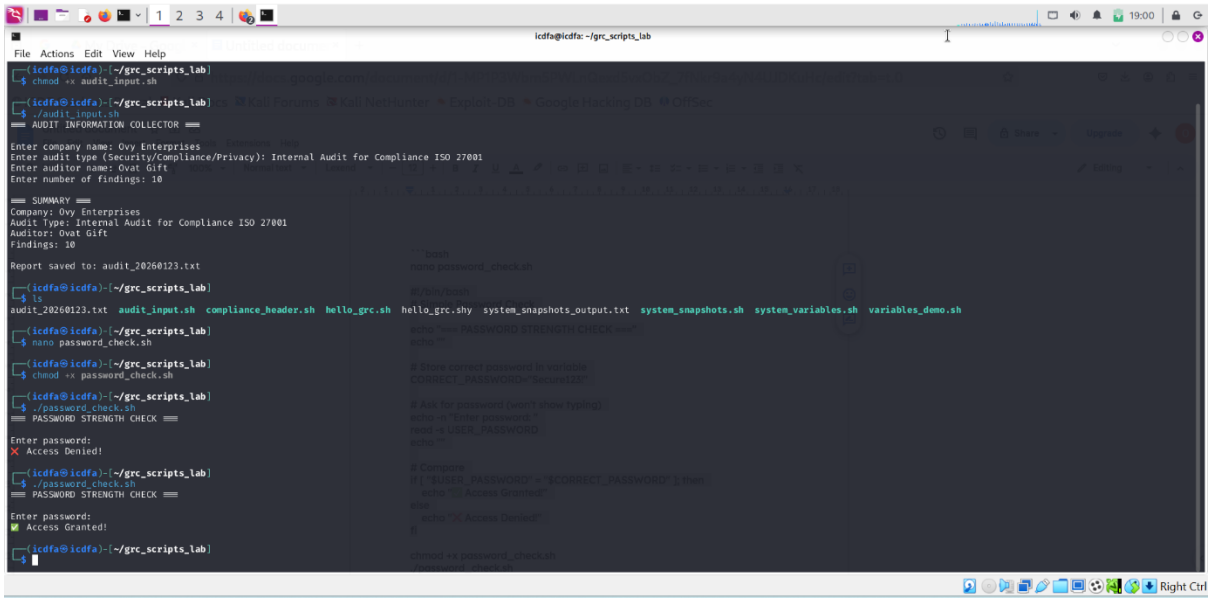
SESSION 2: VARIABLES & INPUT - Storing and Using Data

a) I learned how to declare, assign, and reference variables in Bash.
b) I practised how to use variables to store compliance-related metadata, such as:

- Organisation identifiers
- Audit standards
- Report names

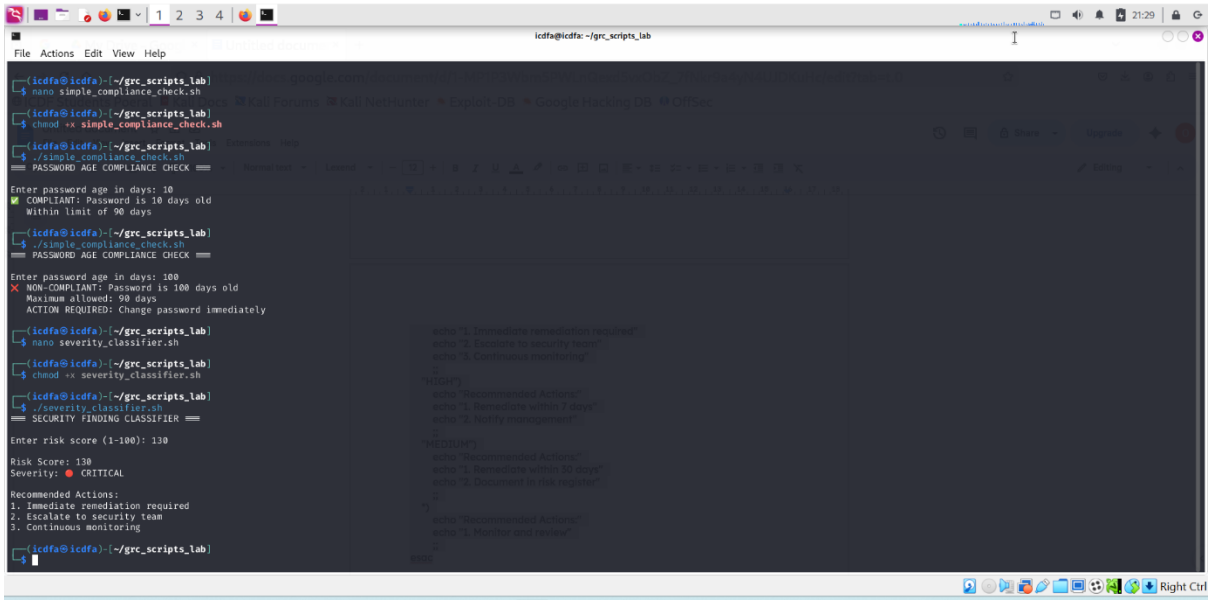
c) I captured system-generated data to support audit evidence and traceability.
d) I implemented secure input handling using `read -s` for sensitive information.

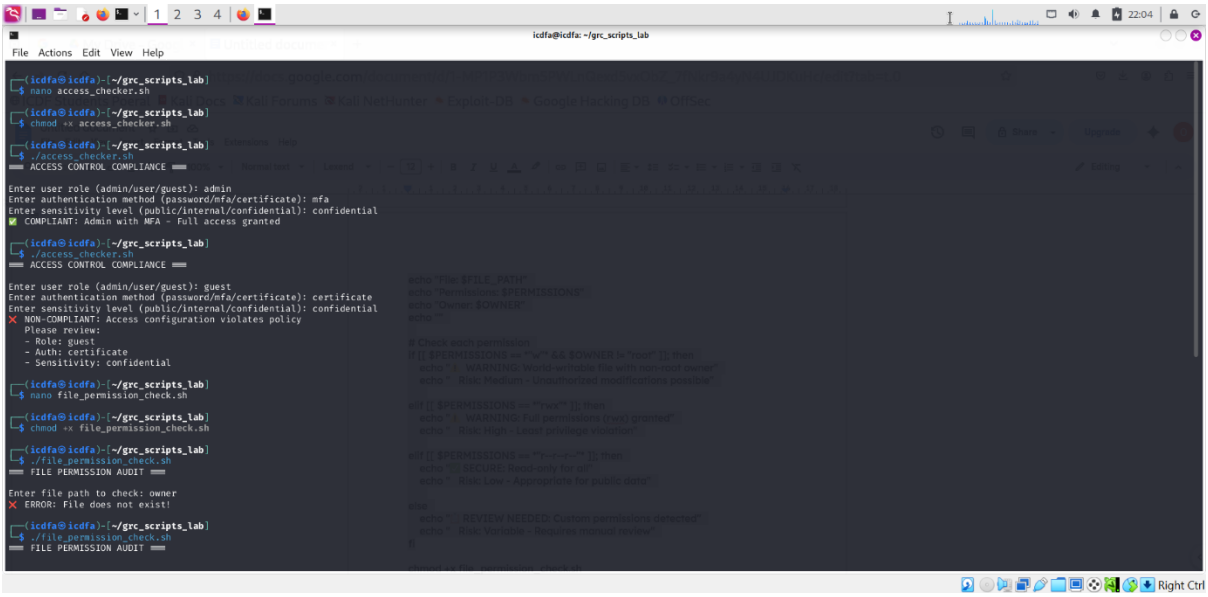




SESSION 3: CONDITIONALS IF/ELSE STATEMENTS - Making Decisions

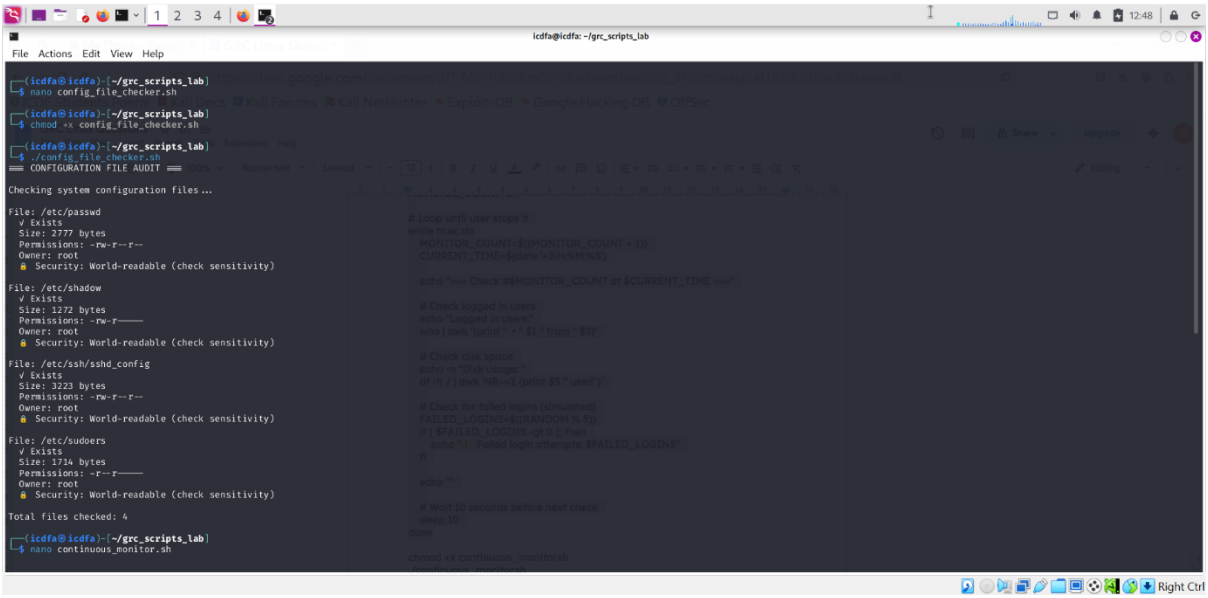
a) I practised how to use the *if/else* and *elif* statements to evaluate system compliance.
b) I applied comparison operators to enforce password policy requirements.
c) I practised how to automate password age validation against defined compliance thresholds.
d) I learned about nested conditionals and practised how to use them to:
    - Perform automated risk assessments
    - Assign severity levels
    - Generate recommended actions
e) This session helped me learn and practice how to convert written security policies into executable logic.

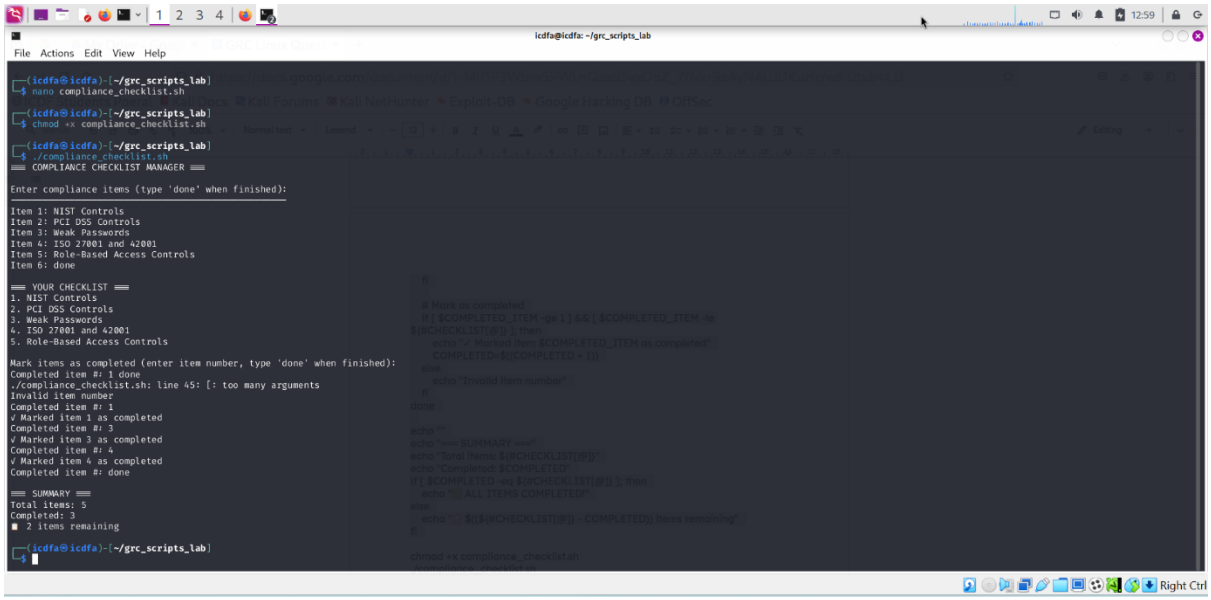




SESSION 4: LOOPS - Repeating Tasks

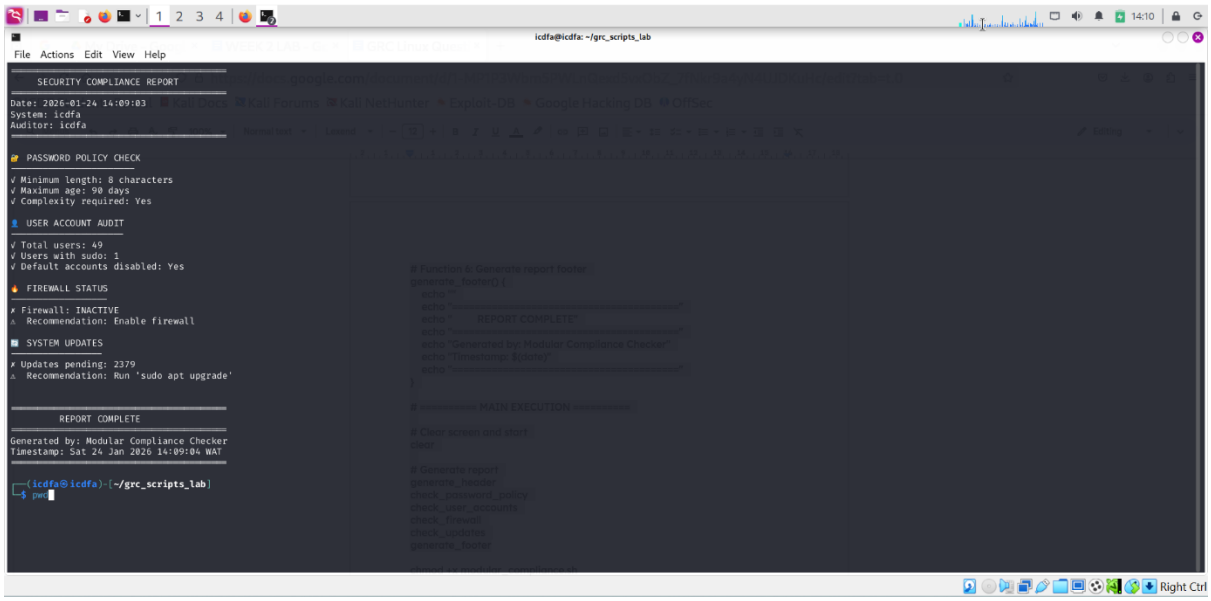
This session focused on loops and the different types of looping mechanisms for batch processing and continuous monitoring through the practical exercises. The exercises gave me an understanding of how to use for loops for the implementation of user account auditing, checks for security compliance, using while loops with timing controls for security monitoring, and simulating real-time compliance oversight. These patterns proved essential for transforming manual, repetitive compliance checks into automated, scheduled processes.

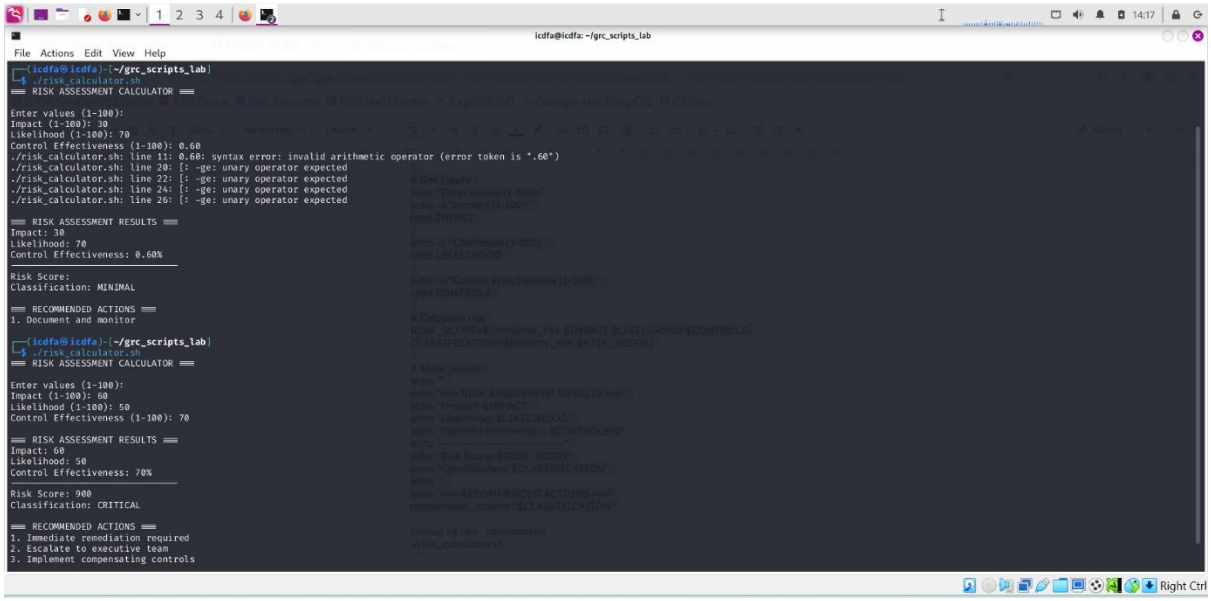


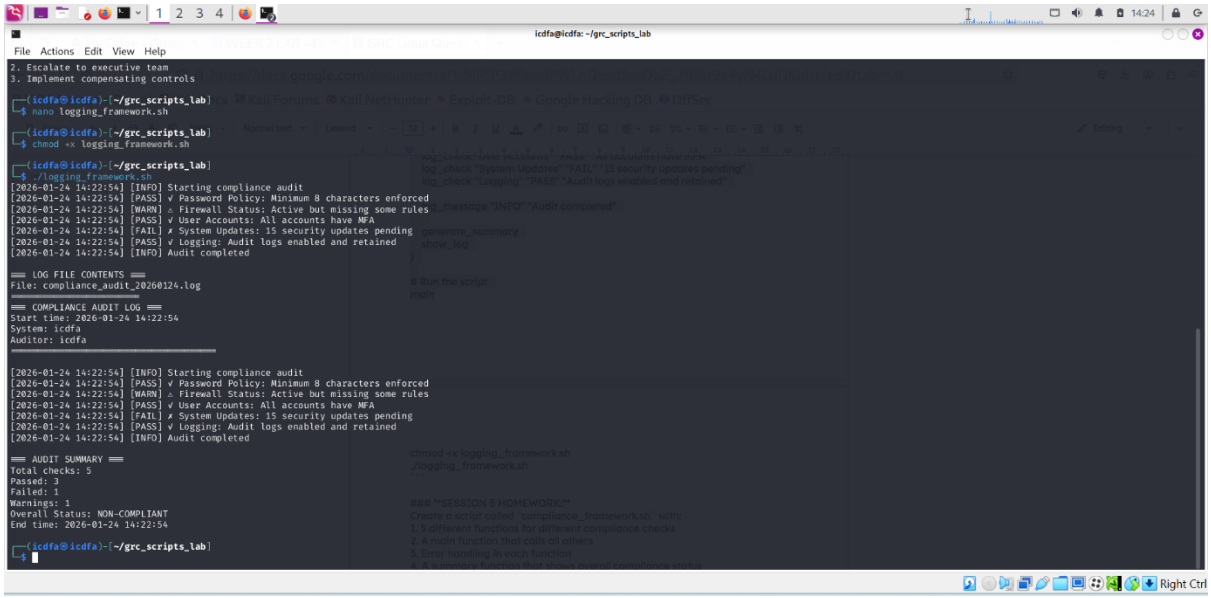


SESSION 5: FUNCTIONS - Organising Code

I understood functions and their uses and focused on code organisation and reusability. I learned how functions can be used for compliance checks, including password policies, user accounts, firewall status, and system updates. I also learned how risk assessment calculations are performed and how to implement mathematical formulas for risk scoring using functions. These exercises gave me insight into how coded and maintained for reuse.

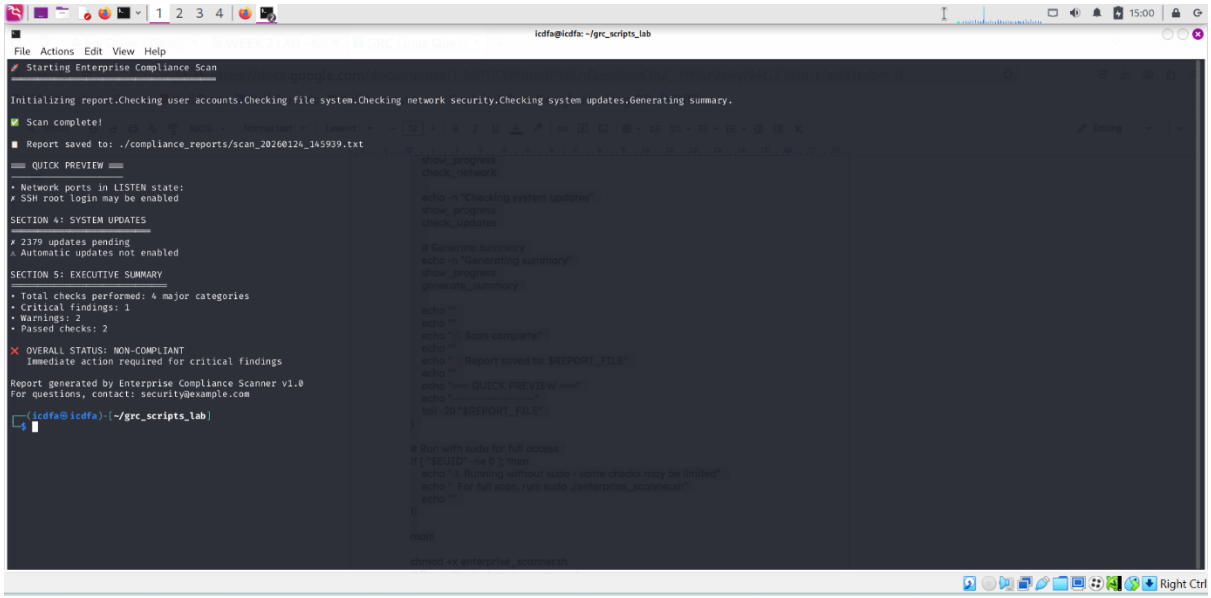


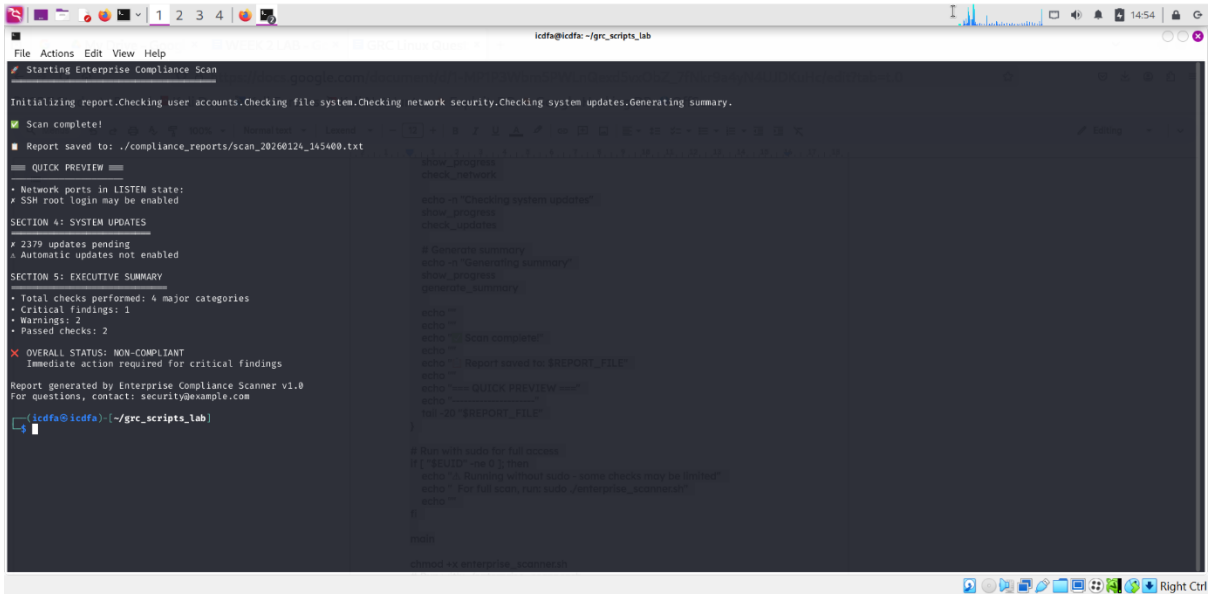




SESSION 6: REAL PROJECTS - Practical Applications

This final session integrated all the concepts into comprehensive enterprise solutions through the enterprise_scanner.sh exercise. This project implemented a complete compliance scanning framework with four major check categories: user account analysis, file system security validation, network configuration auditing, and system update management. The exercise demonstrated practical integration of technical scanning capabilities with business reporting requirements and automated report generation. It demonstrated how scripts can be created for GRC activities.

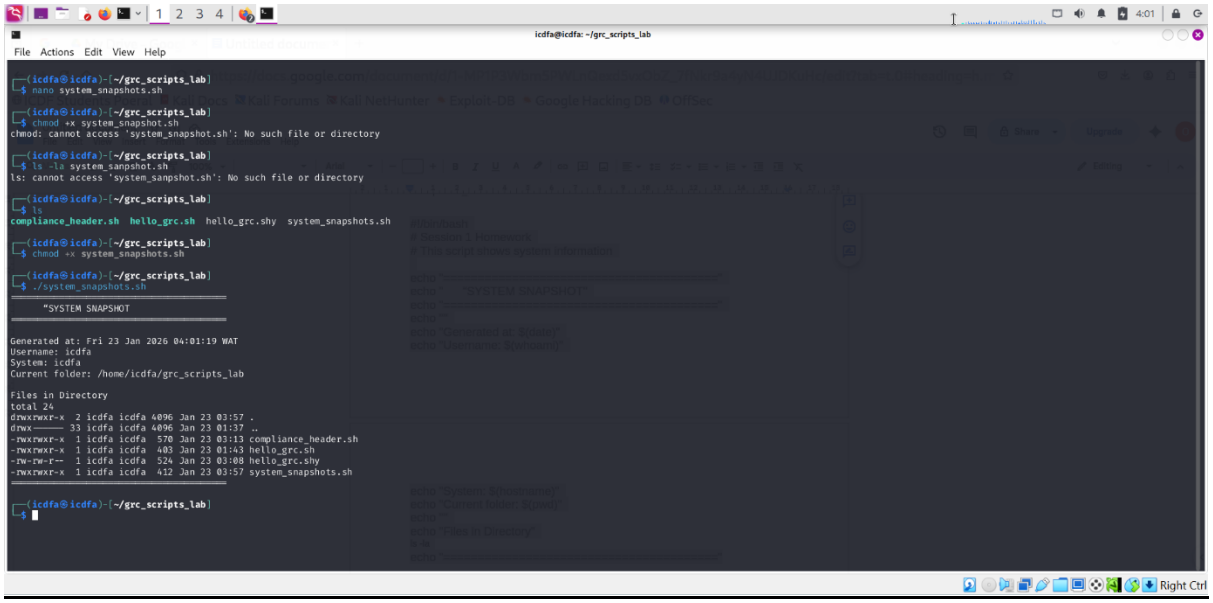


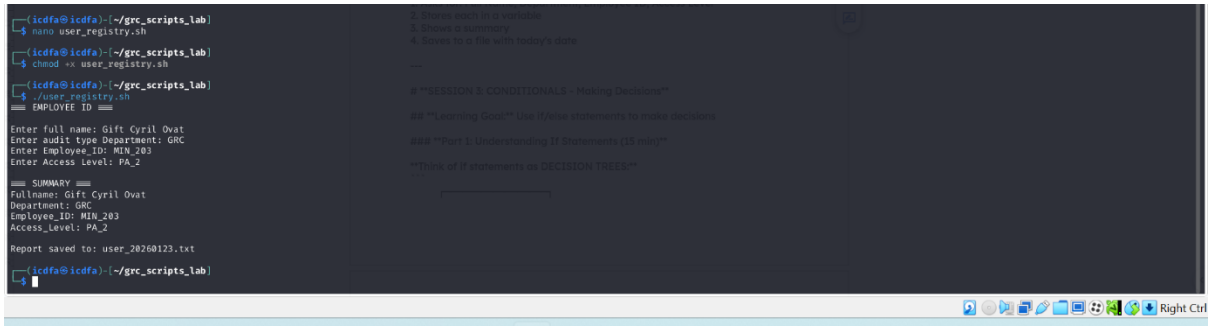


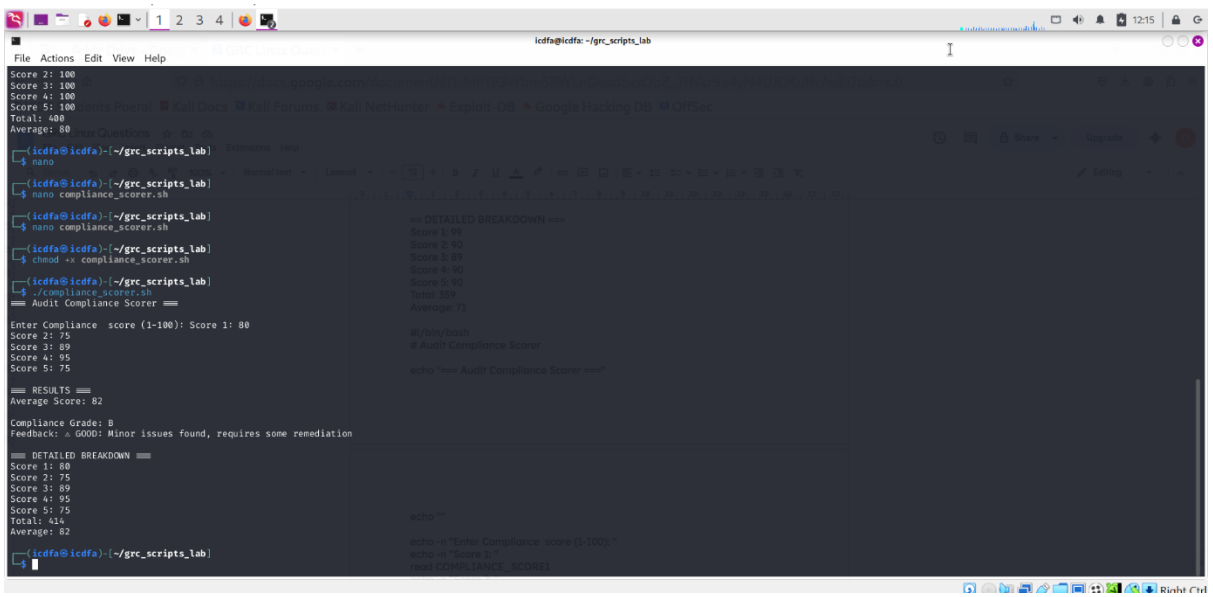
Conclusion

The progressive, exercise-based approach allowed me to practice and understand basic scripting fundamentals for GRC professionals. Through these sessions, I developed comprehensive skills in bash scripting fundamentals, variable use and application, decision logic implementation, automation patterns, and modular code development. This has helped me to understand the of the labs applicability to organisational compliance programs, providing automated solutions for policy enforcement, risk assessment, audit reporting, and continuous monitoring.

Appendixes


icdfa@icdfa: ~/grc_scripts_lab
File Actions Edit View Help
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ nano system_snapshots.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ chmod +x system_snapshot.sh
chmod: cannot access 'system_snapshot.sh': No such file or directory
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ ls -la system_sanpshot.sh
ls: cannot access 'system_sanpshot.sh': No such file or directory
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ ls
compliance_header.sh hello_grc.sh hello_grc.shy system_snapshots.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ chmod +x system_snapshots.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ ./system_snapshots.sh
"SYSTEM SNAPSHOT
Generated at: Fri 23 Jan 2026 04:01:19 WAT
Username: icdfa
System: icdfa
Current folder: /home/icdfa/grc_scripts_lab
Files in Directory
total 24
drwxrwxr-x 2 icdfa icdfa 4096 Jan 23 03:57 .
drwx------ 33 icdfa icdfa 4096 Jan 23 01:37 ..
-rwxrwxr-x 1 icdfa icdfa 570 Jan 23 03:13 compliance_header.sh
-rwxrwxr-x 1 icdfa icdfa 403 Jan 23 01:43 hello_grc.sh
-rw-rw-r-- 1 icdfa icdfa 524 Jan 23 03:08 hello_grc.shy
-rwxrwxr-x 1 icdfa icdfa 412 Jan 23 03:57 system_snapshots.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$
Right Ctrl


┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ nano user_registry.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ chmod +x user_registry.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ ./user_registry.sh
EMPLOYEE ID
Enter full name: Gift Cyril Ovat
Enter audit type Department: GRC
Enter Employee_ID: MIN_203
Enter Access Level: PA_2
SUMMARY
Fullname: Gift Cyril Ovat
Department: GRC
Employee_ID: MIN_203
Access_Level: PA_2
Report saved to: user_20260123.txt
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$
Right Ctrl


icdfa@icdfa: ~/grc_scripts_lab
File Actions Edit View Help
Score 2: 100
Score 3: 100
Score 4: 100
Score 5: 100
Total: 400
Average: 80
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ nano
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ nano compliance_scorer.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ nano compliance_scorer.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ chmod +x compliance_scorer.sh
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$ ./compliance_scorer.sh
Audit Compliance Scorer
Enter Compliance score (1-100): Score 1: 80
Score 2: 75
Score 3: 89
Score 4: 95
Score 5: 75
RESULTS
Average Score: 82
Compliance Grade: B
Feedback: ⚠ GOOD: Minor issues found, requires some remediation
DETAILED BREAKDOWN
Score 1: 80
Score 2: 75
Score 3: 89
Score 4: 95
Score 5: 75
Total: 414
Average: 82
┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$
Right Ctrl

```
icdfa@icdfa: ~/grc_scripts_lab
File  Actions  Edit  View  Help
    Details:
    • Port unexpectedly open
    • Outdated version detected

Service: VPN
  ⚠ LOW RISK - Minor configuration issue
    Details:
    • Port unexpectedly open
    • Outdated version detected
    • Insecure configuration found

Service: API Gateway
  ⚠ LOW RISK - Minor configuration issue
    Details:
    • Insecure configuration found

Service: Load Balancer
  ✅ SECURE - No vulnerabilities found

═══ SCAN ROUND #1 COMPLETE ═══
Timestamp: 2026-01-24 13:58:43
Services scanned: 10
Log entry added to: vulnerability_scan.log

Press Enter to run another scan, or type 'stop' to exit
Your choice: stop

═══ SCANNING STOPPED BY USER ═══
[2026-01-24 13:59:41] Scanning stopped by user after 1 round(s)

Total scan rounds: 1
Log file: vulnerability_scan.log
Vulnerability scan completed!

═══ LOG FILE SUMMARY ═══
Last 5 entries from log:
[2026-01-24 13:58:43] Service: Email - MEDIUM RISK - Multiple issues
[2026-01-24 13:58:43] Service: VPN - LOW RISK - Minor issue detected
[2026-01-24 13:58:43] Service: API Gateway - LOW RISK - Minor issue detected
[2026-01-24 13:58:43] Service: Load Balancer - SECURE - No vulnerabilities
[2026-01-24 13:59:41] Scanning stopped by user after 1 round(s)

Total lines in log: 13

┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$
```

```
icdfa@icdfa: ~/grc_scripts_lab
File  Actions  Edit  View  Help
🚀 STARTING COMPREHENSIVE COMPLIANCE FRAMEWORK
═══════════════════════════════════════

🔥 CHECKING FIREWALL STATUS
──────────────────────────
[sudo] password for icdfa:
⚠ WARNING: iptables found but default policy not set to DROP

🔐 CHECKING PASSWORD POLICY
──────────────────────────
⚠ WARNING: PASS_MIN_DAYS is 0 (should be ≥1)
⚠ WARNING: PASS_MAX_DAYS is 99999 (should be ≤90)
✅ PASS_WARN_AGE: 7 days
⚠ WARNING: Password policy has 2 minor issues

📁 CHECKING CRITICAL FILE PERMISSIONS
──────────────────────────
✅ /etc/sudoers: 440 (correct)
✅ /etc/passwd: 644 (correct)
✅ /etc/shadow: 640 (correct)
✅ /etc/group: 644 (correct)
✅ PASS: All critical files have correct permissions

👤 CHECKING USER ACCOUNTS
──────────────────────────
⚠ WARNING: 28 users without password
✅ No duplicate UIDs found
⚠ WARNING: Root shell is /usr/bin/zsh
⚠ WARNING: User accounts have 2 minor issues

📝 CHECKING LOGGING CONFIGURATION
──────────────────────────
⚠ WARNING: auditd is not installed
✅ rsyslog service is running
✅ Authentication logs are being generated
✅ logrotate configuration exists
⚠ WARNING: Logging has 1 minor issues

═══════════════════════════════════
        COMPLIANCE CHECK SUMMARY
═══════════════════════════════════
Date: 2026-01-24 14:45:29
System: icdfa
Auditor: icdfa
───────────────────────────────────
```

```
icdfa@icdfa: ~/grc_scripts_lab
File  Actions  Edit  View  Help
✅ No duplicate UIDs found
⚠ WARNING: Root shell is /usr/bin/zsh
⚠ WARNING: User accounts have 2 minor issues

📝 CHECKING LOGGING CONFIGURATION
──────────────────────────
⚠ WARNING: auditd is not installed
✅ rsyslog service is running
✅ Authentication logs are being generated
✅ logrotate configuration exists
⚠ WARNING: Logging has 1 minor issues

═══════════════════════════════════
        COMPLIANCE CHECK SUMMARY
═══════════════════════════════════
Date: 2026-01-24 14:45:29
System: icdfa
Auditor: icdfa
───────────────────────────────────

📊 CHECK RESULTS:
──────────────────────────
⚠ Firewall Status: WARNING - iptables found but default policy not set to DROP
⚠ Password Policy: WARNING - Password policy has 2 minor issues
✅ File Permissions: PASS
⚠ User Accounts: WARNING - User accounts have 2 minor issues
⚠ Logging Configuration: WARNING - Logging has 1 minor issues

📈 STATISTICS:
──────────────────────────
Total Checks: 5
Passed: 1
Failed: 0
Warnings: 4

📋 COMPLIANCE SCORE: 20%

⚠ OVERALL STATUS: PARTIALLY COMPLIANT
4 warnings found - review recommended

═══════════════════════════════════
          AUDIT COMPLETE
═══════════════════════════════════

┌──(icdfa㉿icdfa)-[~/grc_scripts_lab]
└─$
```

See links for the lab txt files.

- https://drive.google.com/drive/folders/150Y9v_aTmyFjGJMTkyb3q_rNaiBISD5E?usp=drive_link